ESE 650 Ch.4 Cheatsheet — Rigid-Body Transforms, Quaternions, Mapping & NeRF

SO(3) — Rotation Matrices

$SO(3) := \{R \in \mathbb{R}^{3\times 3} : R^\top R = I,\ \det R = 1\}$, **3 dof**.

Properties: $R^{-1} = R^\top$; $(R_1R_2)^{-1} = R_2^\top R_1^\top$; $\det R = 1$.

Skew-symmetric matrix (hat map $\hat\omega : \mathbb{R}^3 \to \mathfrak{so}(3)$):

$$[\omega]_\times = \hat\omega = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix}, \quad \hat\omega v = \omega \times v$$

Key: $\hat\omega^3 = -\hat\omega$ (so Taylor series closes).

Euler Angles (ZYX)

$R = R_z(\alpha)R_y(\beta)R_x(\gamma)$ (yaw–pitch–roll)

Elementary rotations ($R = R_zR_yR_x$):

$$R_z(\alpha) = \begin{bmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad R_y(\beta) = \begin{bmatrix} \cos\beta & 0 & \sin\beta \\ 0 & 1 & 0 \\ -\sin\beta & 0 & \cos\beta \end{bmatrix}$$

$$R_x(\gamma) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\gamma & -\sin\gamma \\ 0 & \sin\gamma & \cos\gamma \end{bmatrix}$$

Recover from $R = [r_{ij}]$:

$$\alpha = \operatorname{atan2}(r_{21}, r_{11}), \quad \gamma = \operatorname{atan2}(r_{32}, r_{33})$$

$$\beta = \operatorname{atan2}(-r_{31}, \sqrt{r_{32}^2 + r_{33}^2})$$

Gimbal lock: $\beta = \pm 90 \Rightarrow$ yaw and roll become indistinguishable (lose 1 dof). Singularity of Euler angle parametrization.

Rodrigues' Formula

Rotate by angle θ about **unit** axis $\hat\omega$:

$$\boxed{R = I + \sin\theta\,\hat\omega + (1-\cos\theta)\,\hat\omega^2}$$

Equivalently: $R = e^{\hat\omega\theta}$ (matrix exponential).

Derivation key: $\hat\omega^3 = -\hat\omega \Rightarrow$ Taylor series of $e^{\hat\omega\theta}$ splits into sin/cos terms.

Log map (inverse): given R,

$$\theta = \cos^{-1}\left(\frac{\operatorname{tr}(R)-1}{2}\right), \qquad \hat\omega = \frac{R - R^\top}{2\sin\theta}$$

Rotation kinematics:

$$\dot R = \hat\omega R \text{ (world frame)}, \qquad \dot R = R\hat\omega' \text{ (body frame)}$$

where $\omega = R\omega'$ converts body $\leftrightarrow$ world angular velocity.

Lie Groups & Algebra

Lie algebra $\mathfrak{so}(3)$: skew-symmetric matrices $\cong \mathbb{R}^3$.

$$\exp : \mathfrak{so}(3) \to SO(3) \quad \text{(Rodrigues)}, \qquad \log : SO(3) \to \mathfrak{so}(3)$$

SE(3) — rigid body transform:

$$T = \begin{bmatrix} R & t \\ 0 & 1 \end{bmatrix} \in SE(3), \quad T^{-1} = \begin{bmatrix} R^\top & -R^\top t \\ 0 & 1 \end{bmatrix}$$

Transform point: $p' = Rp + t$ (or Tp in homogeneous coords).

Lie algebra $\mathfrak{se}(3)$: $\xi = (\omega, \delta) \in \mathbb{R}^6$:

$$\exp\hat\xi = \begin{bmatrix} e^{\hat\omega} & J_l(\omega)\delta \\ 0 & 1 \end{bmatrix}$$

Quaternions

Unit quaternion $q = (u_0, \mathbf{u})$, $|q| = 1$, represents rotation by θ about $\hat\omega$:

$$\boxed{u_0 = \cos\frac{\theta}{2}, \quad \mathbf{u} = \hat\omega \sin\frac{\theta}{2}}$$

Inverse/conjugate: $q^{-1} = q^* = (u_0, -\mathbf{u})$

Multiplication (Hamilton product):

$$q_1 \otimes q_2 = (u_0v_0 - \mathbf{u}^\top\mathbf{v},\ u_0\mathbf{v} + v_0\mathbf{u} + \mathbf{u}\times\mathbf{v})$$

Non-commutative: $q_1 \otimes q_2 \neq q_2 \otimes q_1$ in general. Note $q \equiv -q$.

Rotate point $x \in \mathbb{R}^3$:

$$q \otimes (0, x) \otimes q^* = (0,\ R(q)\,x)$$

Quaternion to rotation matrix:

$$R(q) = \begin{bmatrix} 2(u_0^2+u_1^2)-1 & 2(u_1u_2-u_0u_3) & 2(u_1u_3+u_0u_2) \\ 2(u_1u_2+u_0u_3) & 2(u_0^2+u_2^2)-1 & 2(u_2u_3-u_0u_1) \\ 2(u_1u_3-u_0u_2) & 2(u_2u_3+u_0u_1) & 2(u_0^2+u_3^2)-1 \end{bmatrix}$$

Rotation matrix $\to$ quaternion: given $R = [r_{ij}]$,

$$u_0 = \tfrac{1}{2}\sqrt{r_{11} + r_{22} + r_{33} + 1}$$

$$u_1 = \frac{r_{32} - r_{23}}{4u_0}, \quad u_2 = \frac{r_{13} - r_{31}}{4u_0}, \quad u_3 = \frac{r_{21} - r_{12}}{4u_0}$$

(valid when $u_0 \neq 0$; use alternative formulas at $\theta = 180$.)

Kinematics:

$$\dot q = \tfrac{1}{2}(0, \omega) \otimes q \text{ (world)}, \qquad \dot q = \tfrac{1}{2}\, q \otimes (0, \omega') \text{ (body)}$$

Why quaternions? No gimbal lock, compact (4 params), numerically stable, easy composition. Preferred over Euler angles for state estimation.

Occupancy Grids

Assumptions: binary cells $m_i \in \{0,1\}$; static world; cells independent.

Log-Odds Filter

Define log-odds: $l(m_i) = \log \frac{p(m_i)}{1-p(m_i)}$

Bayes update for one cell given new obs. y_k, pose x_k:

$$p(m_i \mid y_{1:k}, x_{1:k}) = \frac{p(y_k \mid m_i, x_k)\, p(m_i \mid y_{1:k-1}, x_{1:k-1})}{p(y_k \mid x_k)}$$

Taking log-odds of both sides and using the static-world assumption:

$$\frac{p(m_i \mid y_{1:k})}{1-p(m_i \mid y_{1:k})} = \frac{p(m_i \mid y_k, x_k)}{1-p(m_i \mid y_k, x_k)} \cdot \frac{p(m_i \mid y_{1:k-1})}{1-p(m_i \mid y_{1:k-1})} \cdot \frac{1-p(m_i)}{p(m_i)}$$

Taking log gives the **recursive update** (just addition!):

$$\boxed{l_k(m_i) = \underbrace{l(m_i \mid y_k, x_k)}_{\text{sensor log-odds}} + l_{k-1}(m_i) - \underbrace{l_0(m_i)}_{\text{prior log-odds}}}$$

Recover probability: $p = \sigma(l) = \frac{e^l}{1+e^l}$; occupied iff $l > 0$.

Sensor Models

Sonar/LiDAR along ray to measured distance z:

- Cells *before* z: free (l decreases)
- Cell *at* z: occupied (l increases)
- Cells *beyond* z: unknown (no update)

LiDAR: high-accuracy narrow spike. Sonar: wider, noisier (≈ 15 FOV).

3D: Octrees

Naive 3D grid $\sim$2 billion cells (8 GB) — intractable. **Octree**: recursively subdivide only where needed.

- Empty regions $\to$ coarse (collapsed node)
- Occupied regions $\to$ fine subdivision
- Same log-odds update at leaf nodes

SDF (Signed Distance Function): $\varphi(x)$ = signed distance to surface; surface $= \{x : \varphi(x) = 0\}$. Implicit, handles topology changes.

NeRF & SLAM

Plenoptic Function

The **plenoptic function** $\varphi(x, \lambda) \in \mathbb{R}_+$ gives the radiance seen from viewpoint $x \in SE(3)$ at wavelength λ. It is a **5D function** (position 3D + viewing direction 2D, with roll fixed to zero).

Captures everything a camera can observe at any pose. Too high-dimensional to store directly $\Rightarrow$ approximate with a neural network (**NeRF**).

NeRF = neural network $\xi : SE(3) \to (\sigma(x), c(x))$

σ: volume density; c: RGB color.

Volumetric Rendering

Ray from camera, sample N points $s_1, \ldots, s_N$:

$$\hat{C} = \sum_{i=1}^{N} c(s_i)\, \alpha_i \prod_{j<i} (1-\alpha_j)$$

where $\alpha_i = 1 - e^{-\sigma(s_i)\delta_i}$ (opacity), $\delta_i = s_{i+1} - s_i$.

$\prod_{j<i}(1-\alpha_j)$: transmittance (prob. ray reaches s_i).

Positional encoding: map inputs through Fourier features before MLP to capture high-frequency details:

$$\varphi(x) = \left(\sin(2^k x), \cos(2^k x)\right)_{k=0}^{L-1}$$

SLAM with NeRF

Joint optimization over map ξ and poses $x_{1:k} \in SE(3)$:

$$\hat{\xi}, \hat{x}_{1:k} = \arg\min_{\xi, x_{1:k}} \frac{1}{k} \sum_k \frac{1}{2N} \sum_{i,j} \lambda_k^{ij} (\tilde{y}_k^{ij} - y_k^{ij})^2 - \log \lambda_k^{ij}$$

Challenges: huge data (30 FPS $\times$ 4MP); poses on $SE(3)$ (need structure-aware solvers); expensive backprop through MLP.

Representation Comparison

Rotation	Params	Singular	Use case
Euler angles	3	Yes (gimbal)	Human-readable
Rot. matrix	9	No	Computation
Rodrigues	3	At $\theta = \pi$	Optimization
Quaternion	4	No ($q \equiv -q$)	State estimation

Map type	Pros / Cons
Feature map	Compact; needs landmarks
Occ. grid	Robust; memory heavy in 3D
Octree	Adaptive; LiDAR-friendly
SDF	Implicit; topology-flexible
NeRF	Photo-realistic; slow train